

Status Update 6: Robustness of Contrarian Models on HEARDS Data

Regression, Tree Models, and Feature Pruning Results

Recap of Prior Findings: The Initial Promise

My preliminary analysis indicated that a simple logistic regression model, operating on a contrarian strategy, yielded exceptionally strong results:



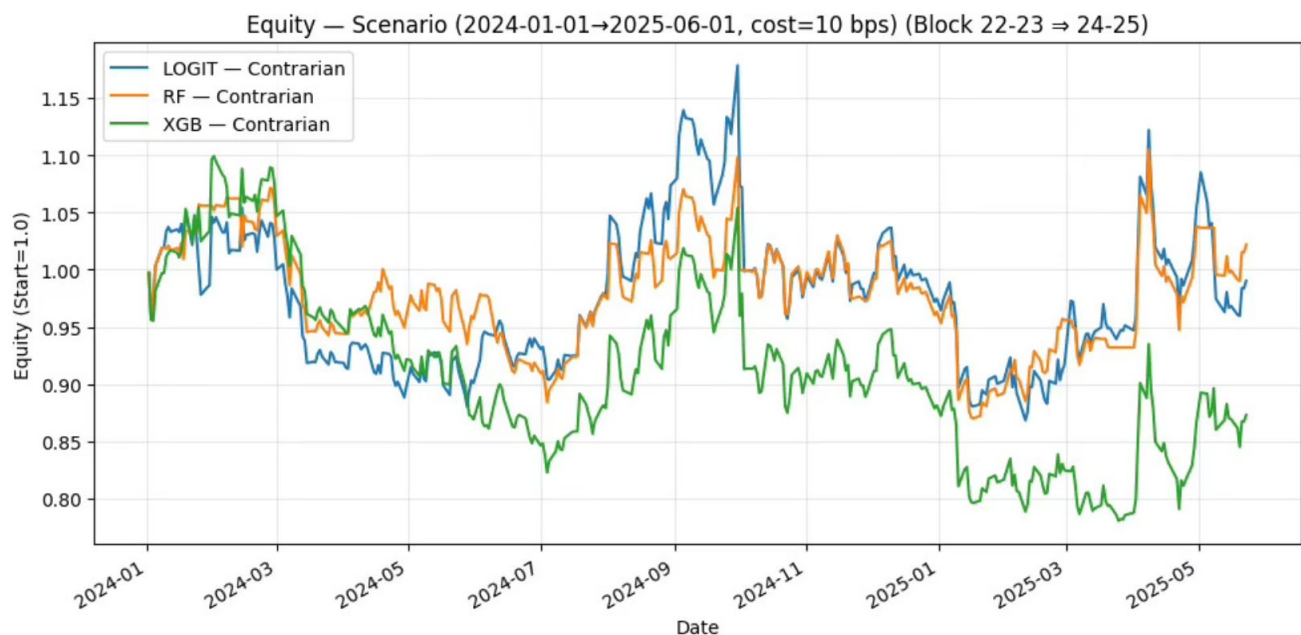
My primary objective is to rigorously compare various model types, systematically prune features, and test the true robustness of these contrarian strategies.

Out-of-Sample Block Split

To rigorously test model robustness and address overfitting concerns, I implemented a strict out-of-sample block split:

Training Period: January 2022 – December 2023

Testing Period: January 2024 – June 2025

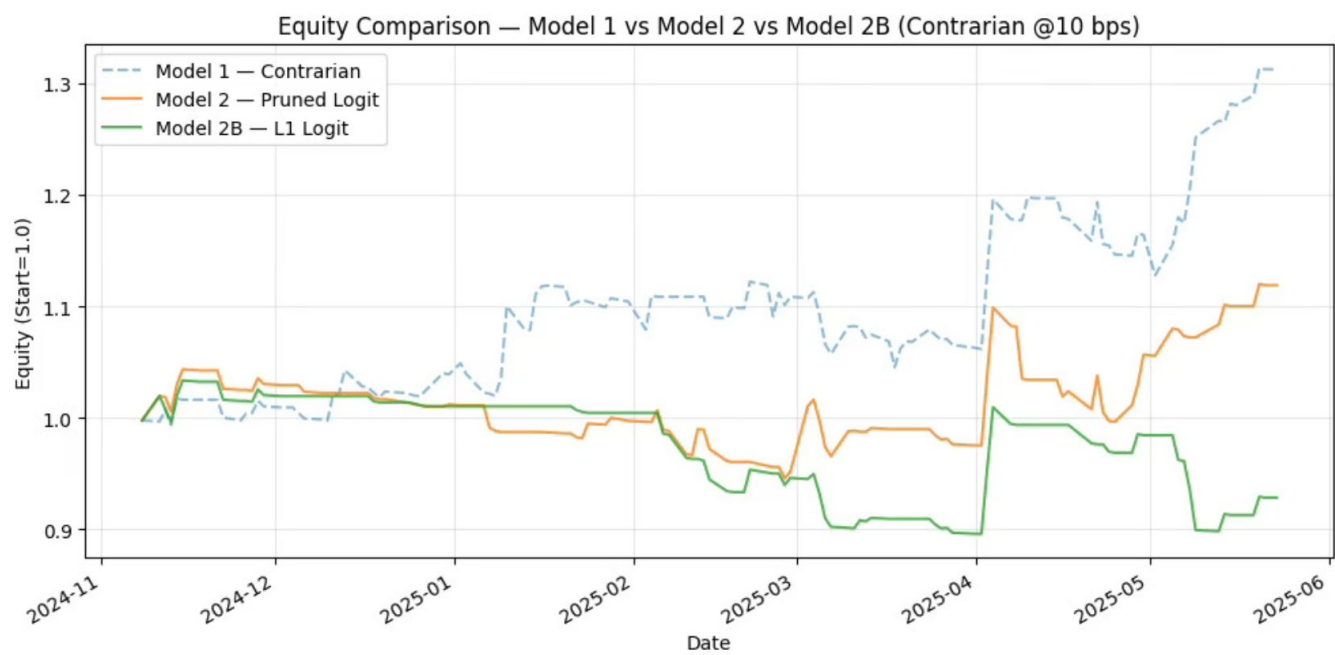


Model	Sharpe	CAGR	MaxDD
Logit	0.12	-0.7%	-26%
Random Forest	0.19	+1.6%	-21%
XGBoost	-0.20	-9.4%	-29%

Under this stringent testing framework, none of the models exhibited consistent positive alpha. This suggests the contrarian effect may be episodic rather than structural.

Re-evaluating Models: Loaded Results (Nov 2024 – Jun 2025)

When examining the performance of "loaded" models in a specific, more favorable period (Nov 2024 – Jun 2025), a clearer hierarchy emerges, particularly for contrarian strategies:

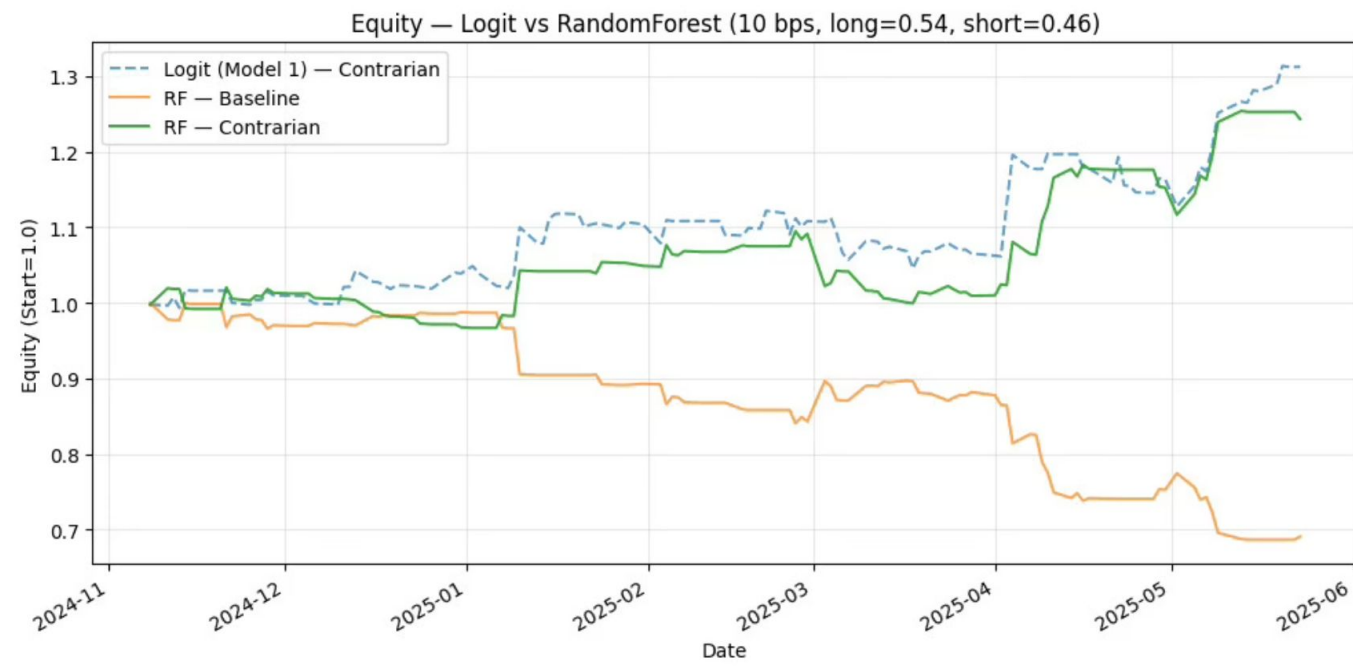


Model	Sharpe Ratio	CAGR	MaxDD
Logit	2.27	67.38%	-6.8%
Random Forest	1.99	51.09%	-8.7%
XGBoost	0.02	-3.68%	-23.3%

The **Logistic model clearly dominates** in this context, demonstrating superior Sharpe and CAGR, with manageable drawdown. This highlights its potential under specific market conditions.

Random Forest: The Contrarian Advantage

My analysis reveals a critical distinction in the performance of Random Forest models based on their strategic approach:



Baseline RF

A standard Random Forest model, without a contrarian overlay, consistently **loses capital**, exhibiting a Sharpe ratio significantly below -2.



Contrarian RF

By contrast, the **Contrarian Random Forest** model achieves a Sharpe ratio of approximately **2.0** and a CAGR of **51%**.

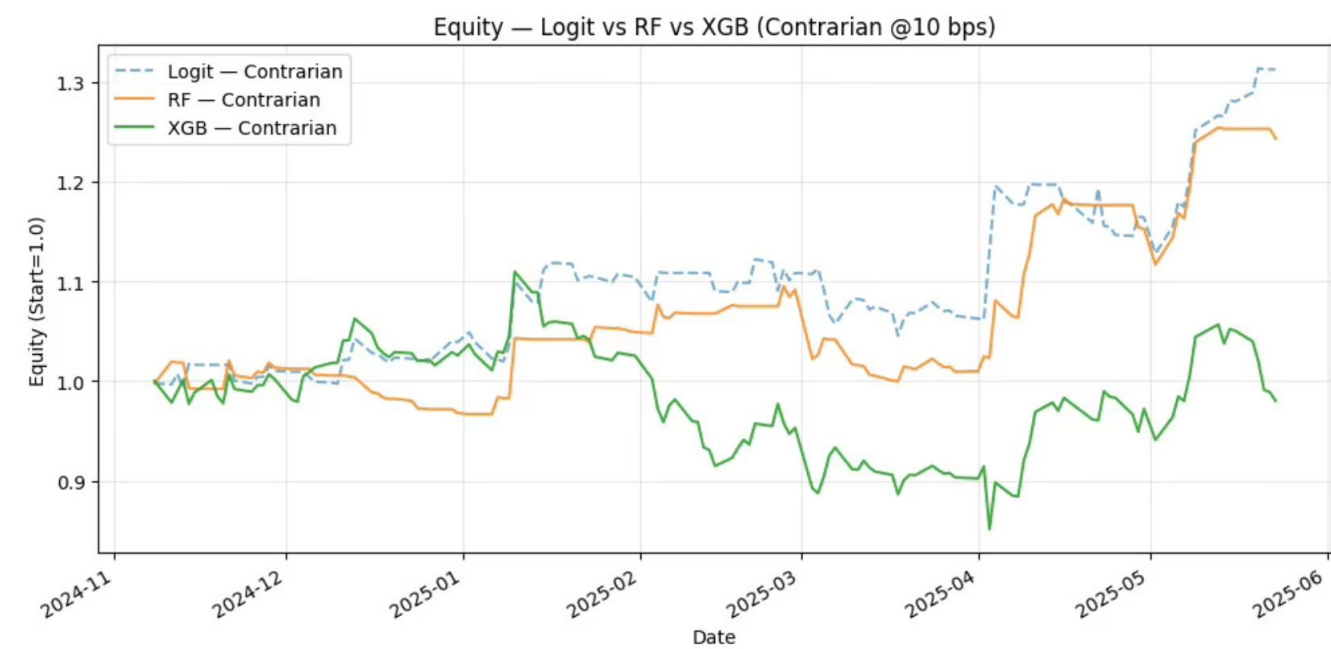
This stark difference underscores that the contrarian flip is crucial for extracting any positive performance from the Random Forest framework on the HEARDS dataset.

Contrarian Models: Head-to-Head Performance

A direct comparison of my primary models, all employing a contrarian execution strategy, solidifies the hierarchy of their effectiveness:

Model (Contrarian)	Sharpe	CAGR
Logistic	2.27	67%
Random Forest	1.99	51%
XGBoost	0.02	-3.7%

The **Logistic model remains the most robust** and consistent performer, followed by the Random Forest. XGBoost, despite its power in other domains, proves to be unstable and ineffective with this strategy and dataset.



Feature Pruning and Regularization

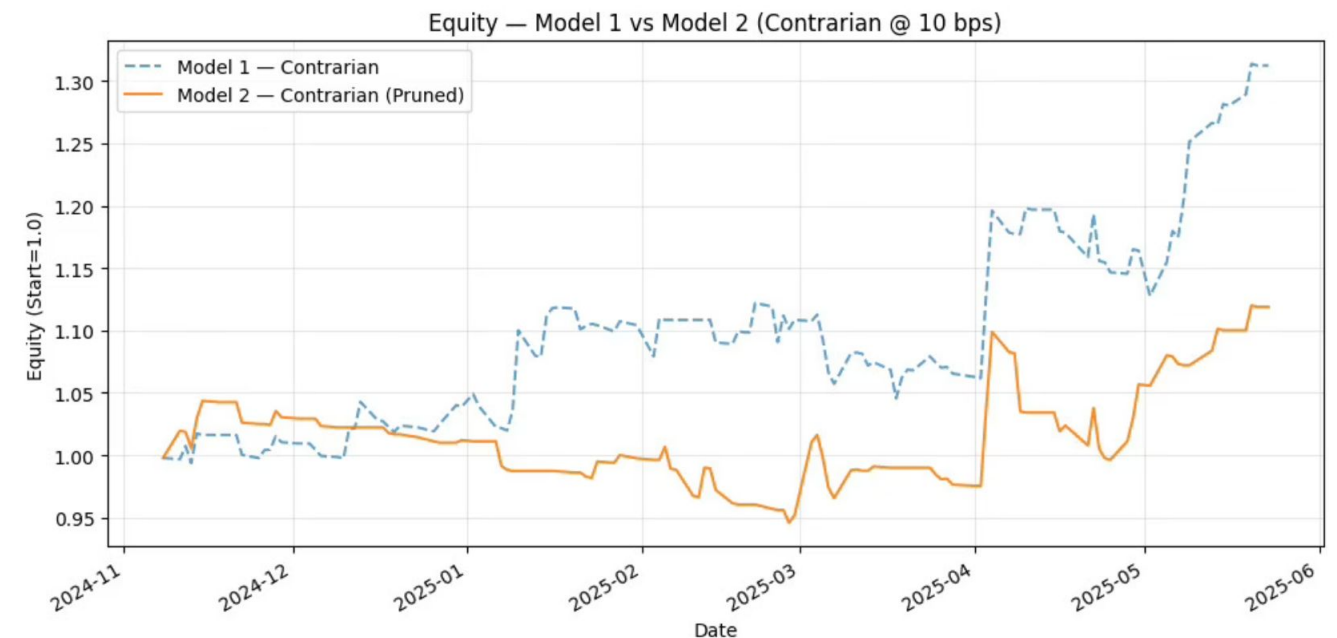
My attempts to simplify models and reduce potential noise through feature pruning and regularization yielded unexpected results:

Pruned Logistic:

- The equity curve of the pruned Logistic model was significantly weaker compared to the full Logistic model.
- Despite identifying and removing highly correlated features, the model lost robustness.

L1 Regularized Logistic:

- The L1-regularized Logistic model consistently underperformed the original, unregularized Logistic model.



This suggests that even features that exhibit high correlation with others still contribute unique predictive value within the context of the contrarian anomaly on HEARDS data.

Summary & Next Steps: Refining Our Search for Alpha



Logistic Model: The Core

The Logistic contrarian model consistently proves to be strongest baseline, exhibiting the most reliable performance characteristics.



Random Forest: Secondary Potential

The Random Forest contrarian model demonstrates some positive edge, although it's less stable than Logistic regression.



XGBoost: Not a Fit

XGBoost fails to add value on this dataset, reinforcing its unsuitability for this particular anomaly.

Robustness checks (block splitting) indicate that the observed anomaly is episodic, not a persistent structural feature of the market.

Next Steps:

- Regime Conditioning: Investigate model performance specifically during high-volatility or other defined market regimes.
- Hybrid Models: Explore combining HEARDS data with broader market features to enhance predictive power and robustness.